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July 24, 1978
SUPERSEDING
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April 15, 1976

FEDERAL SPECIFICATION

RAG, WIPING, COTTON AND COTTON-SYNTHETIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements for cotton wiping rags and cotton-synthetic blended wiping rags made from unused or reclaimed fabrics for use in the wiping of water, oil, and grease from miscellaneous items (see 6.1).

1.2 Classification.

1.2.1 Grades. Wiping rags shall be of the following grades, as specified (see 6.2).

Grade A - 2.0 to 6.0 ounces.
Grade B - 2.0 to 12.0 ounces.

1.2.2 Colors. Wiping rags shall be furnished in the following colors as specified, and shall be applicable to the grades indicated (see 6.2 and 6.3).

White - Grade A or Grade B.
Mixed colors - Grade A or Grade B.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard:

Fed. Test Method Std. No. 191 - Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. Wiping rags shall be of all cotton or cotton-synthetic blends and may consist of unused fabrics from either clean mill ends and mill remnants, or reclaimed fabrics principally from household articles and garments. Rags shall be soft and absorbent and of woven or knitted construction. Heavily napped fabrics, mesh fabrics constructed from hard twisted yarns, and starched or stiffened fabrics will not be acceptable. Rags made from United States flags, National flags of foreign countries or remnants thereof are strictly prohibited.

3.1.1 Defective characteristics. Fabrics submitted for use as wiping rags containing any of the following characteristics shall be considered defective:

(a) Rags containing stains which cover more than one-tenth of the area of the rag.
(b) Rags containing an aggregate of more than 1 square inch of hardened surface.
(c) Rags containing dirt, dust, abrasives, or other clearly noticeable nonfibrous materials.
(d) Rags made of mesh fabrics, such as lace, scrim, and netting except that mesh will be acceptable as a border if not more than 1-inch in depth on an otherwise acceptable rag.
(e) Rags badly worn or tendered as to be unsuited for the intended purpose (see 4.2.7).
(f) Tattered parts less than 3 inches in width and more than 6 inches in length (see 6.4.2).
(g) Edges frayed or raveled continuously or intermittently to a depth of over 2 inches.
(h) Rags made on unopened sleeves, pants, and drawers legs, and tubular fabrics.
(i) Any rags made from United States flags, National flags of foreign countries or remnants thereof.
(j) Rags containing buttons, hooks, eyes, pins, or any other metallic of plastic fittings.
(k) Rags containing elastic yarns amounting to more than 5 percent of the area of the rag.

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(l) Grade A (only) - Rags made from fabrics of which the crotch, cuffs, cuff hems, hems, waistbands, pockets, reinforcements, collars, weltings, and pipings over 1/2 inch in width that have not been completely removed. However, hems in bed sheets and pillowcases 3 inches wide and under will be acceptable provided hems do not have any seam openings totaling more than 4 inches in length.
(m) Any rag not containing cotton fiber.
(n) Grade B (only) - Rags, including those made from pants, overalls, jackets and coats weighing over 9.0 ounces per square yard having pockets, reinforcements (including collar, cuffs, bottonhole and waistband reinforcements), welting and piping (such as used on slip covers, etc.) which have not been completely removed. Also, hems over 1 inch in width, unopened hems and patches with an area in excess of 4 square inches are not acceptable.

3.1.2 The material offered shall contain not less than 85 percent of recovered materials. Recovered materials are defined as material which has been collected or recovered from solid waste. Solid waste is defined as any garbage, refuse, sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility and other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities, but does not include solid or dissolved material in domestic sewage, or solid or dissolved materials in irrigation return flows or industrial discharges.

3.2 Size and measurement. Each wiping rag shall have an area of not less than 288 square inches, and shall be not less than 9 inches wide and not more than 44 inches long. Rags weighing more than 5 ounces with an area of from 225 to 288 square inches will be acceptable when their minimum dimension is not less than 10 inches (see figure 1).

3.3 Weight.

3.3.1 Grade A. Grade A wiping rags shall weigh not less than 2.0 more than 6.0 ounces per square yard (see 3.9).

3.3.2 Grade B. Grade B wiping rags shall weigh not less than 2 ounces nor more than 12 ounces per square yard. At least 35 percent of the net weight of each bale shall be composed of rags weighing not less than 2.0 ounces nor more than 5.0 ounces per square yard (see 3.9).

3.4 Absorbency rate. All rags shall absorb water and oil within 30 seconds when tested in accordance with 4.2.3.

3.5 Washing and sterilization. All rags shall be thoroughly washed, rinsed, and sterilized. (This processing shall be done within the United States, its possessions or Puerto Rico) (see 6.2.1 (a) and 6.7).

3.6 Moisture content. Rags shall have a moisture content no greater than 6.5 percent when tested in accordance with 4.2.4. Moisture content exceeding 10 percent shall be cause for rejection.

3.6.1 Where the reported moisture content for a lot exceeds 6.5 percent, but does not exceed 10 percent, payment adjustment in favor of the purchaser for the entire lot shall be made as follows:

Deduction = Total gross weight of shipment multiplied by percentage of excess moisture content multiplied by price per-pound.

3.7 Put-up. Unless otherwise specified rags shall be put-up in compressed bales of 50 pounds gross weight and with a volume of not more than 3-1/2 cubic feet. Bales shall be permitted a plus or minus tolerance of 3 pounds. Weight variations of more than a plus or minus 3 pounds will be cause for rejection (see 4.2.1).

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3.8 Tare. The allowable average tare weight for each bale shall not exceed 1.75 pounds. Average tare exceeding 1.75 pounds per bale shall be cause for rejection (see 4.2.2).

3.9 Certificates of compliance. A certificate of compliance for each shipment is required for the following to verify that; (1) grade A rags weigh between 2.0 and 6.0 ounces (2) 35 percent of each bale or box of grade B rags is composed of rags weighing between 2.0 and 5.0 ounces, (3) all rags have been sterilized. Where certificates of compliance or reports of analysis are submitted, the Government reserves the right to check test such items to determine the validity of the certificate or analysis.

3.10 Workmanship. Rags shall conform to the quality and grade of product established by this specification and shall be free of defects which may affect their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Weight verification. For inspection and examination of the gross weight and tare weight verification, the lot size shall be expressed in units of bales comprising the inspection lot. The test sample unit shall be one bale using special inspection level S-2. The same sample bales may be used for both tests, but it is important to protect the identity of the individual bales as these examinations and tests are performed (see 4.2.1 and 4.2.2).

4.2.1 Examination of bales for quantity verification (weight). From each inspection lot, a sample of bales shall be selected in accordance with 4.2 for weighing, to verify the average gross weight of each bale. The verified average gross weight of the sample bales shall be multiplied by the number of bales to arrive at the total gross weight of the lot (see 3.7).

4.2.2 Examination of tare weight. The average tare weight of the bales taken for sampling (see 4.2), multiplied by the number of bales in the lot will equal total tare weight. If the total tare weight of the bales exceeds and maximum tare specified in 3.8 the lot shall be rejected.

4.2.3 Test for absorbency. The lot size shall be expressed in units of rags comprising the inspection lot. The test sample unit shall consist of one rag. The sample rags shall be randomly selected using special inspection level S-3 with an AQL of 6.5 percent defective. Not that each bale contains approximately 250 rags. Each sample rag shall be laid on a non-absorbent flat surface. One drop of distilled water and one drop of light machine oil, each 0.04 ml in volume and 72°F $\pm$ 1° shall be applied separately to the applicable sample rags through a capillary by allowing the drops to fall freely onto the rag from a height of 1/2 to 2-inches. After a time interval of 30 seconds, the sample shall be examined and the absorption or non-absorption of the water and oil reported. The rag is considered non-absorbent if the water or oil is not absorbed but remains wholly or partly above the surface of the rag.

4.2.4 Test for moisture content. From the inspection sample (see 4.2), five rags shall be selected randomly from each bale with the moisture content being determined immediately. If this is not possible, the samples, immediately upon selection, shall be placed as a composite unit into an air tight container(s) and sealed to prevent moisture loss. The sealed container(s) is then weighed (this weight is the original weight, Wo), the lid of the container is removed and

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the container is immediately placed in the oven at 221°F to 230°F (105°C to 110°C) for a minimum of 2 hours. Immediately upon removing the container(s) from the oven, the container(s) shall be resealed and allowed to cool. After the container has cooled, the sealed container is weighed (this weight is the dry weight, Wd).

The moisture content in percentage form can be calculated using the following formulation applying the values for Wo and Wd:

$$\text{Moisture content percent} = \frac{Wo-Wd}{Wo} \times 100$$

The moisture content of the samples shall be considered the moisture content of the entire lot. If the moisture content of the sample unit exceeds 7.5 percent, the entire lot shall be rejected (see 3.6).

4.2.5 Test for cotton fiber content. The lot size shall be expressed in units of rags comprising the inspection lot. The test sample unit shall consist of one rag. The sample rags shall be randomly selected using special inspection level S-3 with an AQL of 1.5 percent defective. Note that each bale contains approximately 250 rags. Rags chosen for the absorbency test may be used. The rags shall be tested in accordance with paragraph 4.2.5.1 to determine the presence of cotton fiber (see 3.1).

4.2.5.1 Identification of cotton. To remove oils, waxes, and dirt, the specimen shall be washed in one of the following: ether, acetone, alcohol or a 5 percent aqueous solution of sodium hydroxide. To remove dyes, one of the following reagents shall be used.

(1) By oxidation, using nitric acid, hydrogen peroxide, or chlorine water.
(2) By solution, using alchol, acetic acid, hydrochloric acid, or pyridine
(3) By reduction, using hydrosulphite, stanners chloride, or hydrochloric acid with metallic zinc. After dirt, oils, waxes, and dyes have been removed place a dozen or so fibers from the sample onto a slide. The fibers shall be immersed in three or four drops of Herzbergs stain: covered with a slide avoiding air bubbles, allowed to stand 2 mins with surplus solution drained off. Examine with microscope using transmitted light and 100 diameters. Magnification up to 500 diameters may be used if a more detailed examination is necessary. The sample fibers shall be compared with a reference sample which has been treated in the same manner.

Identification. The color of cotton (Gossypium sp.) in the natural state, ranges from white to ecru. The staple length of commercial cottons varies from about 1/4 inch to slightly more than 2 inches. Microscopically, most fibers appear much like a twisted ribbon with rounded edges. They have no longitudinal or cross marking; the lumens vary from very narrow to over two-thirds of the diameter of the fibers. Diameters of the fibers vary from about 9 to 25 microns and average from 16 to 20 microns. Undyed cotton fibers are stained pink to dark red in color by Herzberg's stain.

4.2.6 Examination for visual and dimensional characteristics. For the purpose of visual and dimensional inspection, the sample unit shall be one rag. The number of bales selected for the inspection and test above (see 4.2) shall be multiplied by 250 (approx number of rags per bale) to determine the number of individual rags in the inspection lot. Sampling shall be in accordance with inspection level II of MIL-STD-105 and an AQL of 2.5 for majors and 4.0 for minors expressed as percent defective. Rags shall be selected at random with no more than 50 rags coming from any single inspection bale or box. During this examination, if any rag is suspected of not containing cotton (see table II) by visual examination or by feel, the rag(s) is to be tested in accordance with 4.2.5. If the rag(s) fails the test in accordance with 4.2.5 it shall be classified as a major defect.

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TABLE I. Examination for visual and dimensional characteristics

Characteristics	Defects	Classification Major	Classification Minor
	NOTE One or more rags made from United States flags, National flags of foreign countries, or remnants thereof (see 3.1) shall cause the entire lot to be rejected.		
Grades A and B	a. Rags containing no cotton fiber. (Any amount of linen, flax fiber is acceptable).	X	
	b. Dirt, dust, and abrasive material.		X
	c. Rags containing elastic yarns amounting to more than 5 percent of the area of the rag.		X
	d. Objectionable odor.		X
	e. Rags not white (see 6.3.1) when specified in contract or order.		X
	f. Not of specified dimensions (see 3.2).	X	
	g. An aggregate of more than 1 square inch of hardened surface.	X	
	h. Rags weighing less than or more than amount specified (see 3.3).	X	
	i. Rags made of mesh fabrics, such as laces, scrim, and netting, except that mesh will be acceptable as a border not more than 1 inch depth on an otherwise acceptable rag.	X	
	j. Rags made from starched or stiffened fabrics.	X	
	k. Rags made from heavy napped fabrics and fabrics woven with hand twisted yarn.	X	
	l. Stains which cover more than one-tenth of the area of the rag.	X	
	m. Badly worn or tendered as to be unsuited for the intended purpose.	X	
	n. Tattered parts less than 3 inches wide and more than 6 inches long.		X
	o. Edge frayed or raveled continuously or intermittently to a depth of over 2 inches.		X
	p. Unopened sleeves, pants, and drawers legs, and tubular fabrics.		X
	q. Buttons, hooks, closed safety pins, eyes or any other metallic or plastic fittings (one or more).		X
	r. Rags that have not been thoroughly washed, rinsed, and sterilized.	X	
	s. Rags containing fiber content labels that declare 100 percent synthetic fiber.	X	
	t. Foreign objects of dangerous materials; such as nails, tacks, needles, pins, glass, sharp edged metal items, etc.	1/ 3X	
Grade A (only)	a. Rags or patches thereto that are other than white.	X	
	b. Rags made from bed sheets and pillow cases:		
	1. Containing hems more than 3 inches wide.	X	
	2. Open seams along the hems totaling more than 4 inches in length.		X

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TABLE I. Examination for visual and dimensional characteristics (Cont'd)

Characteristics	Defects	Classification Major	Classification Minor
Grade A (only)	c. Rags, except "b" above, containing crotches, cuffs, cuff hems, hems, waistbands, pockets, collars, reinforcements, weltings, and pipings over 1/2 inch in width.	X	
Grade B (only)	a. Rags from pants, overalls, coveralls, jackets, and coats 7.0 to 9.0 ounces inclusive, containing crotches, cuffs, cuff hems, hems, pockets, and waistbands over 1 inch in width.	X	
	b. Rags, including those from items in "a" above, over 9.0 ounces containing pockets, reinforcements, welting, and piping. Unopened hems over 1 inch and patches over 4 square inches.	X	

1/ Defect(t) is equivalent to 3 major defects as signified by "3X".

4.2.7 Examination for tenderness. Wiping rags which are in question as to tenderness (see 3.1.1(e) shall be examined in the following manner:

The rag shall be grasped on opposite edges so that it remains fully opened and flat. There shall be approximately 3 inches of material gathered in each hand. With the arms extended, the rag shall be subjected to a steady outward force. If the rag pulls apart, it shall be considered tender.

4.2.8 Certification. A certificate shall be submitted with each bid specifying that the material to be supplied will contain the percentage of recovered material required herein (see 3.1.2).

4.3 Examination of preparation for delivery. The following examination of packaging and marking requirements of section 5 of this specification shall be in accordance with an AQL of 4.0 expressed in terms of percent defective, and inspection level S-2 of MIL-STD-105.

5. PREPARATION FOR DELIVERY

5.1 Commercial packing. Wiping rags of one grade only shall be furnished in bales in the weight and volume specified in paragraph 3.7. The bales shall be covered with clean serviceable burlap material and secured with strapping, wire ties, or rope. The bales shall be packaged in accordance with normal commercial practice and in a manner that will assure acceptance by common carrier and provide product protection against breakage or loss during multiple shipments, handling, and storage. The shipping unit shall be in compliance with National Motor Freight Classification or Uniform Freight Classification.

5.2 Marking. Marking shall be as specified in the contract or order.

6. NOTES

6.1 Intended use. Rags are primarily used in the wiping away of water; wiping of oil and grease from machinery, and for miscellaneous cleaning.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Grade required (see 1.2.1).

NOTE: It is suggested that white rags, grades A and B be ordered only where better quality rags are required.

(c) Color required (see 1.2.2).

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6.2.1 Purchasers should specify the following contractual requirements:

(a) That a certified copy of any affidavit, by a responsible officer of the contractor's organization, be submitted with each shipment to the effect that all rags furnished have been washed, rinsed sterilized and dried within the United States, its possessions or Puerto Rico (see 3.5). For the purposes of this specification the term sterilized implies that the rags have been submitted to temperatures of not less than 180°F during washing cycles and 250°F during heat drying (exhaust air 175°F).

6.3 Interpretation.

6.3.1 White. The word "white" as contained in 1.2.2 shall be interpreted to mean the following type rags: unbleached (gray) rags that were in the gray state and have been partially bleached or fully bleached, and colored rags which have had the dye color completely removed (known as stripped rags) except that evidence of a dye color on seams of stripped rags will be acceptable. Ornamental color trimmings are acceptable, provided they are not over 5 percent of the area of the rag.

6.3.2 Mixed colors. The words "mixed colors" as used in 1.2.2 is intended to include rags of any color, including "white" rags as acceptable.

6.4 Definitions.

6.4.1 Size and measurements. The width and length dimensions as specified in 3.2 are defined as follows:

The minimum width of a rag shall be the measurement between the two closest opposing points on the perimeter of a rag. The maximum length of a rag shall be the distance between the most distant points along the edges of a rag. When rags are nearly rectangular, the longer side shall be considered the "length".

6.4.2 Tatter. A tatter is a fabric part protruding from the body of the fabric.

6.4.3 Tubular fabric. A tubular fabric is a woven or knitted fabric which is seamless, such as pillow tubing or circular knitted fabric.

6.5 Basis of purchase. Wiping rags shall be purchased by gross weight, in accordance with requirements of 3.6 and 3.8 and determinations in 4.2.1.

6.7 In the process of rinsing rags, 3.5, it is permissible to use a wetting-out agent to increase the absorbency of rags provided that rags meet all requirements of this specification, and are nontoxic.

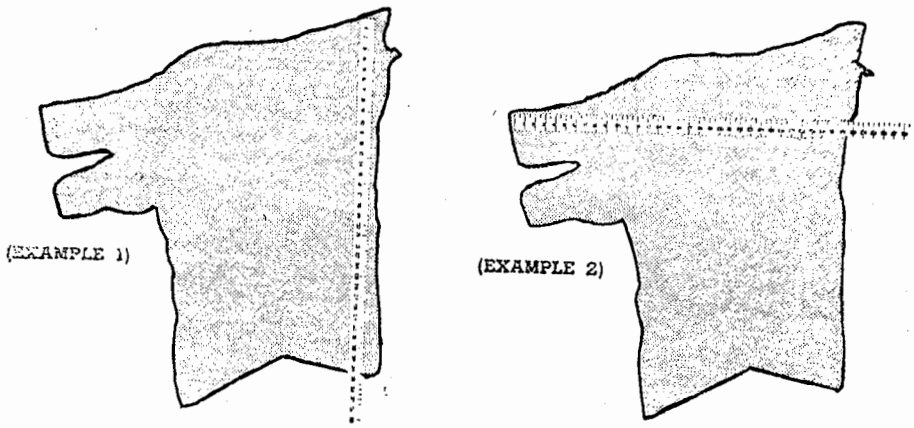
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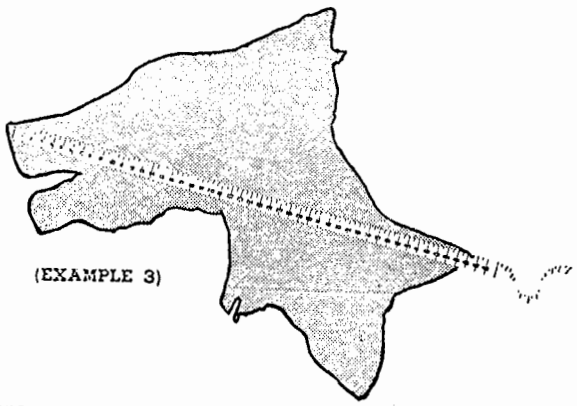
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ACCEPTABLE METHOD FOR MEASURING WIPING RAGS



UNACCEPTABLE METHOD FOR MEASURING WIPING RAGS

Figure 1 — Rags shall be smoothed and laid on a flat surface when measurements are taken.

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